

As well-known,  $(\mathbb{R}, +, \cdot)$  is the Complete ordered field.  
 Consider a set  $\mathbb{R}^2$ . Define two operations on  $\mathbb{R}^2$  as:

$$(x_1, y_1) + (x_2, y_2) \stackrel{\text{def}}{=} (x_1 + x_2, y_1 + y_2)$$

$$(x_1, y_1) \cdot (x_2, y_2) \stackrel{\text{def}}{=} (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2)$$

operation on  $\mathbb{R}^2$

defined on  $\mathbb{R}$ .

$$(x, y)^{-1} =$$

Then, the triple  $(\mathbb{R}^2, +, \cdot)$  is a field:

$$\begin{cases} \text{Identity} & (0, 0) \\ \text{Unity} & (1, 0) \\ \text{inverse} & (x, y)^{-1} = \left( \frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right) \end{cases}$$

Def. Define a Norm of  $(a, b) \in \mathbb{C}$  as:

$$\|(a, b)\| \stackrel{\text{def}}{=} \sqrt{a^2 + b^2}.$$

Lemma.  $\|z\|^2 = z \cdot \bar{z}$ . Moreover, if  $z \neq 0$ , then  $z^{-1} = \frac{\bar{z}}{\|z\|^2}$

$$\begin{aligned} (i h) \cdot (-i h) \\ = -i^2 \cdot h^2 = h^2. \end{aligned}$$

Consider a Complex map:

$$f: \Omega \rightarrow \mathbb{C} : (x, y) \mapsto (u(x, y), v(x, y))$$

Where  $\Omega \subseteq \mathbb{C}$ . Set-Theoretically, it equals to a map from a subset of  $\mathbb{R}^2$  into  $\mathbb{R}^2$ .  
 Moreover, Topologically,  $\mathbb{C}$  is a Metric space induced by  $\|\cdot\|$ , homeomorphic to  $\mathbb{R}^2$ .

Therefore, the continuity of  $f$  is not different whether we see that  $\mathbb{C}$  as  $\mathbb{C}$  or  $\mathbb{R}^2$ .  
 But, in the Differentiation, Complex and  $\mathbb{R}^2$  have crucial difference. Recall that

$f: A(\subset \mathbb{R}^m) \rightarrow \mathbb{R}^n$  is differentiable at  $a \in A^\circ$  if and only if there exists a linear  $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$  s.t.

$$\frac{f(a+h) - f(a) - T(h)}{\|h\|_m} \rightarrow 0 \text{ as } h \rightarrow 0.$$

If the derivative  $T$  of  $f$  at  $a \in A^\circ$  exists, it is unique.

Equivalently, if  $f$  has a derivative  $T$  at  $a \in A^\circ$ , then for small enough  $h \in \mathbb{R}^m$ ,

$$f(a+h) - f(a) = \|h\| \cdot \left[ \frac{f(a+h) - f(a) - T(h)}{\|h\|} \right] + T(h)$$

Moreover, if  $f$  is differentiable at  $a \in A^\circ$ , then all directional derivatives of  $f$  at  $a$  exist,

$$f'(a)u = Df(a) \cdot u$$

But, since  $\mathbb{C}$  is a field, we can define derivative without a norm,

this is the first bifurcation that indicates  $\mathbb{C}$  does not equal  $\mathbb{R}^2$ .

Def: Let  $\Omega \subseteq \mathbb{C}$  open, and let  $f: \Omega \rightarrow \mathbb{C}$  be a map.  
 Given  $z_0 \in \Omega$ ,  $f$  is said to be holomorphic at  $z_0$  if: the limit

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} \quad (h \in \mathbb{C}, h \neq 0)$$

exists. The limit is called a derivative of  $f$  at  $z_0$ , denote  $f'(z_0)$ .

Note: Holomorphic = Regular = Complex differentiable  $\stackrel{\text{not definition}}{=} \text{Analytic}$

Cauchy - Riemann equation.

Let  $f: \Omega \rightarrow \mathbb{C}: (x, y) \mapsto (u(x, y), v(x, y))$  be a map where  $\Omega \subseteq \mathbb{C}$  open set.

If  $f$  is holomorphic at  $z_0 = (x_0, y_0)$ , then the partial derivatives of  $u, v$  at  $z_0$  exist and

$$\frac{\partial u}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) \quad \text{and} \quad \frac{\partial u}{\partial y}(z_0) = -\frac{\partial v}{\partial x}(z_0) \quad \dots (*)$$

hold. (Or  $u_x(z_0) = v_y(z_0)$  and  $u_y(z_0) = -v_x(z_0)$ )

Conversely, if  $u$  and  $v$  are continuously differentiable at  $z_0 = (x_0, y_0)$  satisfying (\*), then  $f$  is holomorphic at  $z_0$ . More specifically,

$$f'(z_0) = \frac{\partial u}{\partial x}(z_0) + i \cdot \frac{\partial v}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0) - i \cdot \frac{\partial u}{\partial y}(z_0)$$

Proof. Suppose that  $f$  is holomorphic at  $z_0$ . Then,

$$\begin{aligned} f'(z_0) &= \lim_{h_1 \rightarrow 0} \frac{f(x_0 + h_1, y_0) - f(x_0, y_0)}{h_1} = \lim_{h_2 \rightarrow 0} \frac{f(x_0, y_0 + h_2) - f(x_0, y_0)}{ih_2} \\ &= \frac{\partial f}{\partial x}(z_0) = \frac{1}{i} \cdot \frac{\partial f}{\partial y}(z_0) \end{aligned}$$

Therefore, at  $z_0$ ,  $u_x + i \cdot v_x = v_y - i \cdot u_y$ .

Conversely, suppose that  $u, v$  are continuously differentiable at  $z_0$ . That is,

$u_x, u_y, v_x, v_y$  exist and are continuous at  $z_0$ .

Thm. Define a complex map as

$$f: \mathbb{D}_R \rightarrow \mathbb{C} : z \mapsto \sum_{n=0}^{\infty} a_n z^n$$

where  $R = \left[ \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} \right]^{-1}$ . Then,  $f$  is differentiable by terms on the domain as

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}.$$

Proof. Since  $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$ , we obtain

$$\limsup |a_n|^{\frac{1}{n}} = \limsup |n a_n|^{\frac{1}{n}}$$

Thus  $\sum a_n z^n$  and  $\sum n a_n z^{n-1}$  have same convergence boundary,

Define a map on  $\mathbb{D}_R$  as:

$$g(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

Enough to show that  $g = f'$ . Let  $z_0 \in \mathbb{D}_R$  be fixed, and  $r \in \mathbb{R}$  s.t.  $|z_0| < r < R$ .

Since  $f(z)$  converges, denote  $S_N(z) = \sum_{n=0}^N a_n z^n$  and  $E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n$ . ( $f = S_N + E_N$ ).

For  $h \in \mathbb{C}$  s.t.  $|z_0 + h| < r$ ,

$$\frac{f(z_0 + h) - f(z_0)}{h} - g(z_0) = \left[ \frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) \right] + (S'_N(z_0) - g(z_0)) + \left[ \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right]$$

$$\text{Meanwhile, } \left| \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right| \leq \sum_{n=N+1}^{\infty} |a_n| \cdot \left| \frac{(z_0 + h)^n - z_0^n}{h} \right| \leq \sum_{n=N+1}^{\infty} |a_n| \cdot n r^{n-1} \rightarrow 0 \text{ as } N \rightarrow \infty$$

$$(z_0 + h)^n - z_0^n = (z_0 + h - z_0) \left[ (z_0 + h)^{n-1} + (z_0 + h)^{n-2} z_0 + \dots + z_0^{n-1} \right]$$

And clearly  $S'_N(z_0) - g(z_0) \rightarrow 0$  as  $N \rightarrow \infty$ . And,

$$\left| \frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) \right| \rightarrow 0 \text{ as } h \rightarrow 0.$$

Therefore  $g = f'$  for any  $z_0 \in \mathbb{D}_R$ .

Corollary. Power series defined on convergence disk is  $C^\infty$ .

Curve.

A continuous map  $Z: [a, b] \rightarrow \mathbb{C}; t \mapsto Z(t)$  is called a parametrized Curve.

If it is 'injection', then called Simple,

If  $Z(a) = Z(b)$ , then called closed.

It is called simple closed if injective on  $[a, b)$ .

Moreover, if  $Z$  is continuously differentiable on  $[a, b]$  and  $Z' \neq 0$ , then it is smooth.

A smooth curves  $\begin{cases} Z: [a, b] \rightarrow \mathbb{C} \\ \tilde{Z}: [c, d] \rightarrow \mathbb{C} \end{cases}$  are equivalent if: there is  $t: [c, d] \rightarrow [a, b]$  s.t.  
 $t$  is 1-1,  $t$  is continuously differentiable,  $t' > 0$ ,  $\tilde{Z} = Z \circ t$ .

Integral

Def. Let  $\gamma$  be a smooth curve parametrized by  $Z: [a, b] \rightarrow \mathbb{C}$ ,  
and let  $f: \mathbb{C} \rightarrow \mathbb{C}$  be a continuous map on  $\gamma$ . Define an Integral of  $f$  along  $\gamma$  as:

$$\int_{\gamma} f(z) dz \stackrel{\text{def}}{=} \int_a^b f(Z(t)) \cdot Z'(t) dt.$$

And  $\text{length}(\gamma) = \int_a^b |Z'(t)| dt.$

Thm. Let  $f: \Omega \rightarrow \mathbb{C}$  be a map where  $\Omega \subseteq \mathbb{C}$  is open, and  $\gamma$  be a curve from  $w_1$  to  $w_2$ .  
If  $f$  has primitive function  $F$  on  $\Omega$ , then

$$\int_{\gamma} f(z) dz = F(w_2) - F(w_1).$$

Goursat's Thm. Let  $f: \Omega \rightarrow \mathbb{C}$  be a holomorphic and a triangle  $T \subseteq \Omega$ . Then,

$$\int_T f(z) dz = 0.$$

Lemma. In open disk, a holomorphic map has a primitive map in the disk.

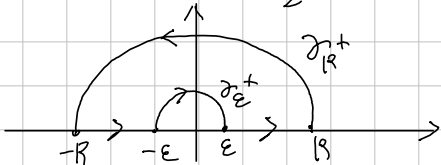
Corollary. If  $f$  is holomorphic in the disk, then for any closed curve  $\gamma$  in the disk,

$$\int_{\gamma} f(z) dz = 0.$$

Calculate

prove  $\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$ .

Proof. Let  $f(z) = \frac{1 - e^{iz}}{z^2}$ . And, set the contour as



Observe that: for any  $0 < \epsilon < R < \infty$ ,

$$\int_{\gamma} f(z) dz = \int_{-R}^{-\epsilon} \frac{1 - e^{iz}}{z^2} dx + \int_{\gamma_{\epsilon}^+} \frac{1 - e^{iz}}{z^2} dz + \int_{\epsilon}^R \frac{1 - e^{iz}}{z^2} dx + \int_{\gamma_R^+} \frac{1 - e^{iz}}{z^2} dz = 0.$$

First,  $\left| \frac{1 - e^{iz}}{z^2} \right| \leq \frac{2}{|z|^2}$  implies  $\left| \int_{\gamma} f(z) dz \right| \leq \sup_{z \in \gamma_R^+} \left| \frac{1}{z^2} \right| \cdot \text{length } \gamma_R^+ = \frac{1}{R^2} \cdot \pi R \rightarrow 0$

as  $R \rightarrow \infty$ . Therefore,  $\int_{|x| \geq \epsilon} \frac{1 - e^{ix}}{x^2} dx = - \int_{\gamma_{\epsilon}^+} \frac{1 - e^{iz}}{z^2} dz$ .

Since  $\gamma_{\epsilon}^-: z(t) = \epsilon \cdot e^{it}$  ( $t \in [0, \pi]$ ), and observe that

$$\frac{1 - e^{iz}}{z^2} = \frac{-zi + zi + 1 - e^{iz}}{z^2} = -\frac{i}{z} + \frac{1 + zi - e^{iz}}{z}$$

By L'Hopital Thm,  $\lim_{z \rightarrow 0} \frac{1 + zi - e^{iz}}{z} = \lim_{z \rightarrow 0} i + ie^{iz} = i + i = 2i$ .

Therefore  $\frac{1 + zi - e^{iz}}{z}$  is bounded in the neighborhood of zero. Take  $M > 0$  as bound.

Now,  $\int_{\gamma_{\epsilon}^-} -\frac{i}{z} dz = \int_0^{\pi} \frac{i}{\epsilon \cdot e^{it}} \cdot \epsilon \cdot i \cdot e^{it} dt = \int_0^{\pi} 1 dt = \pi$

and  $\int_{\gamma_{\epsilon}^-} \frac{1 + zi - e^{iz}}{z} dz \leq M \cdot \pi \epsilon \rightarrow 0$  as  $\epsilon \rightarrow 0$

Finally,  $\int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2} dx = \lim_{\epsilon \rightarrow 0} \int_{|x| \geq \epsilon} f(z) dz = \pi$ .

Since  $\frac{1 - \cos x}{x^2}$  is even,  $\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$ .

# Thm. Cauchy Integral Formula.

Suppose  $f$  is Holomorphic on some open set containing a closure of disk  $D$ .  
If  $C$  is a Curve with Positive orientation of  $\partial D$ , then for any  $z_0 \in D$ ,

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{z - z_0} dz$$

Proof. Let  $z_0 \in D$  be fixed, and consider a Curve  $\Gamma_{\epsilon}$  as  
Since  $\frac{f(z)}{z - z_0}$  is Holomorphic on  $D \setminus \{z_0\}$ , therefore



$$\int_{\Gamma_{\epsilon}} \frac{f(z)}{z - z_0} dz = 0. \text{ Taking } \epsilon \rightarrow 0, \text{ then we obtain}$$

$$\lim_{\epsilon \rightarrow 0} \int_{\Gamma_{\epsilon}} \frac{f(z)}{z - z_0} dz = \int_{C_{\epsilon}} \frac{f(z)}{z - z_0} dz + \int_C \frac{f(z)}{z - z_0} dz = 0.$$

Meanwhile,  $\frac{f(z)}{z - z_0} = \frac{f(z) - f(z_0)}{z - z_0} + \frac{f(z_0)}{z - z_0}$  and since  $f$  is Holomorphic,  
the first term is bounded, therefore  $\lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{f(z) - f(z_0)}{z - z_0} dz = 0.$

$$\begin{aligned} \text{Now, } \lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{f(z_0)}{z - z_0} dz &= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{1}{z - z_0} dz \\ &= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \left( \int_0^{2\pi} \frac{1}{\epsilon \cdot e^{it}} \cdot \epsilon \cdot i \cdot e^{it} dt \right) \\ &= -f(z_0) \cdot 2\pi i. \end{aligned}$$

$$\text{Consequently, } \int_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

# Corollary. $\int_C \frac{f(z)}{(z - z_0)^{n+1}} dz = 2\pi i \cdot \frac{f^{(n)}(z_0)}{n!}.$

Proof. Using induction for  $n$ . For  $n=0$ , clear. If  $f^{(n-1)}(z_0) = \frac{(n-1)!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^n} dz$ , then

$$\begin{aligned} \frac{f^{(n-1)}(z_0 + h) - f^{(n-1)}(z_0)}{h} &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[ \frac{1}{(z - z_0 - h)^n} - \frac{1}{(z - z_0)^n} \right] dz \\ &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[ \frac{h}{(z - z_0 - h)(z - z_0)^n} \cdot \left( \sum_{k=0}^{n-1} \frac{1}{(z - z_0 - h)^{n-1-k}} \cdot \frac{1}{(z - z_0)^k} \right) \right] dz \\ &\rightarrow \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{(z - z_0)^2} \cdot \sum_{k=0}^{n-1} \frac{1}{(z - z_0)^{n-1-k}} dz \quad \text{as } h \rightarrow 0 \\ &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot n \cdot \frac{1}{(z - z_0)^{n+1}} dz \\ &= \frac{n!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz = f^{(n)}(z_0). \end{aligned}$$

# Thm. Cauchy inequality.

$$|f^{(n)}(z_0)| \leq \frac{n!}{R^n} \cdot \sup_C |f| \text{ if } f \text{ is Holomorphic in an open containing } \overline{B_R(z_0)}, C = \partial B_R(z_0).$$

Cauchy's Integral Formula, Revised.

Thm. Let  $f: G \rightarrow \mathbb{C}$  be a Holomorphic on open  $G \subset \mathbb{C}$ .

If  $z_0 \in G$  and  $\overline{B_r(z_0)} \subseteq G$ , then for a Curve  $C: z_0 + re^{it}$  ( $0 \leq t \leq 2\pi$ ),

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{z - z_0} dz.$$

Proof. Denote  $D = B_r(z_0)$ . Let  $z_0 \in D$  be fixed, and consider a Curve  $\gamma_{\delta, \epsilon}$  as

Since  $\frac{f(z)}{z - z_0}$  is Holomorphic on  $D \setminus \{z_0\}$ ,  $\int_{\gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz = 0$ .



Taking  $\delta \rightarrow 0$ , then we obtain

$$\lim_{\delta \rightarrow 0} \int_{\gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz = - \int_{C_\epsilon} \frac{f(z)}{z - z_0} dz + \int_C \frac{f(z)}{z - z_0} dz = 0.$$

Meanwhile,  $\frac{f(z)}{z - z_0} = \frac{f(z) - f(z_0)}{z - z_0} + \frac{f(z_0)}{z - z_0}$  and since  $f$  is Holomorphic,

the first term is bounded, therefore  $\lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{f(z) - f(z_0)}{z - z_0} dz = 0$ .

$$\text{Now, } \lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{f(z_0)}{z - z_0} dz = f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{1}{z - z_0} dz$$

$$= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \left( \int_0^{2\pi} \frac{1}{\epsilon \cdot e^{it}} \cdot \epsilon \cdot i \cdot e^{it} dt \right)$$

$$= f(z_0) \cdot 2\pi i.$$

$$\int_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

Consequently,

$$\int_C \frac{f(z)}{z - z_0} dz = \lim_{\delta \rightarrow 0} \int_{\gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz + \int_{C_\epsilon} \frac{f(z)}{z - z_0} dz = 2\pi i \cdot f(z_0).$$

Corollary. 
$$\int_C \frac{f(z)}{(z - z_0)^{n+1}} dz = 2\pi i \frac{f^{(n)}(z_0)}{n!}$$

Thm. Suppose  $f$  is Holomorphic on an open  $G$  containing a closed disk  $\overline{D}$  having center  $z_0$ ,  

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n \quad (z \in D)$$

Proof. 
$$f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} d\mu \quad \left( \frac{1}{\mu - z} = \frac{1}{\mu - z_0 - (z - z_0)} = \frac{1}{\mu - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\mu - z_0}\right)} \right)$$

$$= \frac{1}{2\pi i} \cdot \int_C f(\mu) \cdot \sum_{n=0}^{\infty} \frac{1}{(\mu - z_0)^{n+1}} \cdot (z - z_0)^n d\mu$$

$$= \sum_{n=0}^{\infty} \left( \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{(\mu - z_0)^{n+1}} d\mu \right) \cdot (z - z_0)^n$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n$$

Conclary, Liouville's Theorem.

If  $f: \mathbb{C} \rightarrow \mathbb{C}$  is entire and bounded, then it is constant.

Proof. Given  $z_0 \in \mathbb{C}$  and  $R > 0$ ,  $|f'(z_0)| \leq \frac{1}{R} \cdot \sup_{C_R} |f| \rightarrow 0$  as  $R \rightarrow \infty$ . Therefore,  $f' = 0$ .  
 Thus  $f$  is constant.

Fundamental Theorem of Algebra.

Non-constant polynomial  $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0 \in \mathbb{C}[z]$  has a root in  $\mathbb{C}$ .

Proof. Suppose that  $P(z)$  has no root in  $\mathbb{C}$ . Then,  $1/P(z)$  is well-defined on  $\mathbb{C}$ , entire. observe that 
$$\frac{P(z)}{z^n} = a_n + \left( \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right) \quad (z \neq 0).$$

The terms  $a_{n-k} \cdot z^{-k}$  ( $k=1, \dots, n$ ) converges to zero as  $|z| \rightarrow \infty$ , thus for some  $R > 0$ ,

$$\begin{aligned} |P(z)| &\geq 2|a_n| \cdot z^n \quad (|z| > R) \\ &\geq 2|a_n| \cdot R^n \end{aligned}$$

Therefore it is bounded below in  $|z| > R$ , and clearly in  $|z| \leq R$ .

Now,  $\frac{1}{P}$  is bounded, thus constant. Contradiction.

Thm. Let  $G$  be a connected open set and let  $f: G \rightarrow \mathbb{C}$  be an analytic function. Then, TFAE:

a)  $f \equiv 0$ .

b) There is a point  $a \in G$  such that  $f^{(n)}(a) = 0$  for each  $n \geq 0$ .

c)  $f^{-1}[\{0\}]$  has a limit point in  $G$ .

Proof. a)  $\Rightarrow$  b) and a)  $\Rightarrow$  c) are trivial. To show c)  $\Rightarrow$  a),

let  $a \in G$  be a limit point of  $Z = f^{-1}[\{0\}]$ . Since  $Z$  is closed,  $a \in Z$ , or  $f(a) = 0$ .

Suppose that for some  $n \in \mathbb{N}$ ,

$$f(a) = f'(a) = \dots = f^{(n-1)}(a) = 0 \text{ but } f^{(n)}(a) \neq 0.$$

Since  $f$  is analytic, 
$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (z-a)^k = \sum_{k=n}^{\infty} \frac{f^{(k)}(a)}{k!} (z-a)^k.$$

in  $|z-a| < R$ , where  $R > 0$  s.t.  $B_R(a) \subset G$ .

Put 
$$g(z) = \sum_{k=n}^{\infty} a_k (z-a)^{k-n} \quad \left( a_k = \frac{f^{(k)}(a)}{k!} \right)$$

then,  $g$  is analytic in  $B_R(a)$ , and  $f = (z-a)^n g$ , and  $g(a) = a_n \neq 0$ .

Now, we can find  $0 < r < R$  s.t.  $g(z) \neq 0$  for  $|z-a| < r$ , but since  $a$  is the limit point of  $Z$ , so for some  $b$  s.t.  $f(b) = 0$  and  $0 < |b-a| < r$ , thus

$$f(b) = 0 = (b-a)^n \cdot g(b) \Rightarrow g(b) = 0,$$

which is Contradiction.

⌈ Singularities.

⌈ Def. A complex function  $f$  has an isolated singularity at  $z=a$  if:

there is an  $R > 0$  st  $f$  is defined and analytic in  $B_R(a) - \{a\}$ , but not in  $B_R(a)$ .

An isolated singularity  $a \in \mathbb{C}$  of  $f$  is called:

- Removable if: there is an analytic extension of  $f$  to  $B_R(a)$ .
- Pole if:  $\lim_{z \rightarrow a} |f(z)| = \infty$ .
- essential if: neither Removable nor Pole.

⌈ Thm. Let  $a \in \mathbb{C}$  be an isolated singularity of  $f$ . Then,

$a \in \mathbb{C}$  is removable singularity iff  $\lim_{z \rightarrow a} (z-a) \cdot f(z) = 0$ .

⌈ Prop. If  $G \subseteq \mathbb{C}$  is a region containing  $a \in \mathbb{C}$  and  $f$  is analytic on  $G \setminus \{a\}$  with a pole  $a$ , then there exists  $m \in \mathbb{N}$  and analytic  $g: G \rightarrow \mathbb{C}$  such that

$$f(z) = \frac{g(z)}{(z-a)^m}$$

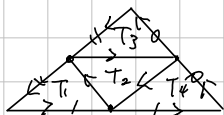
The smallest such a number is called order of a pole.



Cauchy - Goursat's Thm.

Thm. A map  $f$  is holomorphic on open set  $\Omega \subseteq \mathbb{C}$  and a triangle  $T$  is contained in  $\Omega$  then  $\int_T f(z) dz = 0$ .

Proof. Let  $T^{(0)} = T$  with positive oriented, and  $T_1^{(1)}, T_2^{(1)}, T_3^{(1)}, T_4^{(1)}$  be:



Then,  $\int_{T^{(0)}} f(z) dz = \sum_{i=1}^4 \int_{T_i^{(1)}} f(z) dz$ , because the integral on  $T_2$  is canceled.

So, for some  $j \in \{1, 3, 4\}$ ,  $\left| \int_{T^{(0)}} f(z) dz \right| \leq 4 \cdot \left| \int_{T_j^{(1)}} f(z) dz \right|$

Put  $T^{(0)} = T_j^{(1)}$ . Inductively, for each  $n \in \mathbb{N}$ ,

$$\left| \int_{T^{(n)}} f(z) dz \right| \leq 4^n \cdot \left| \int_{T^{(0)}} f(z) dz \right|$$

And, by construction,

$$T^{(0)} \supseteq T^{(1)} \supseteq \dots$$

Def. Let  $G$  be an open set of  $\mathbb{C}$ .

A function  $f: G \rightarrow \mathbb{C}$  is called Analytic if  $f$  is continuously differentiable.

## Logarithm

Def. If  $G \subseteq \mathbb{C}$  is an open connected set in  $\mathbb{C}$  and  $f: G \rightarrow \mathbb{C}$  is a continuous function s.t.  
 $z = \exp f(z)$  for all  $z \in G$

Then,  $f$  is called a branch of the logarithm.

Explicitly, if  $z = x + iy \in G$ , then let  $\log z = a + ib$ . Now,

$$\exp(\log z) = \exp(a) \cdot \exp(ib) = e^a \cdot (\cos b + i \sin b) = x + iy.$$

Thus,  $\begin{cases} e^a \cos b = x \\ e^a \sin b = y \end{cases}, \quad \begin{cases} |e^a (\cos b + i \sin b)| = |e^a| = |z|, \text{ so } a = \ln |z|. \\ b = \arg z. \end{cases}$

Therefore,  $\log z = \ln |z| + i \arg z$ .

Cauchy-Riemann equation.

Let  $f: G \rightarrow \mathbb{C}$  be a map on region  $G$  and let  $U(x,y) = \operatorname{Re} f(x,y)$ ,  $V(x,y) = \operatorname{Im} f(x,y)$ . Suppose that  $U$  and  $V$  have continuous partial derivatives. Then,

$f$  is analytic if and only if  $U$  and  $V$  satisfy the Cauchy-Riemann equation.

Proof Suppose that  $f$  is analytic. For  $h \in \mathbb{R} \setminus \{0\}$ ,

$$\frac{f(z+h) - f(z)}{h} = \frac{U(x+h,y) - U(x,y)}{h} + i \frac{V(x+h,y) - V(x,y)}{h}$$

Letting  $h \rightarrow 0$ , then  $f'(z) = \frac{\partial U}{\partial x}(x,y) + i \frac{\partial V}{\partial x}(x,y)$ .

Meanwhile, for  $h \in \mathbb{R} \setminus \{0\}$ ,

$$\frac{f(z+ih) - f(z)}{i \cdot h} = -i \frac{U(x,y+h) - U(x,y)}{h} + i \frac{V(x,y+h) - V(x,y)}{h}$$

Letting  $h \rightarrow 0$ , then  $f'(z) = -i \frac{\partial U}{\partial y}(x,y) + \frac{\partial V}{\partial y}(x,y)$

So,  $U_x = V_y$  and  $V_x = -U_y$ . (\*) [we using continuously differentiable.]

Conversely,  $U$  and  $V$  satisfy the condition (\*), and  $U_x, U_y, V_x, V_y$  are continuous.

Let  $z = x+iy \in G$  be given. Since  $G$  is open, for some  $r > 0$ ,  $B_r(z) \subseteq G$ .

For  $h = s+it \in B_r(0)$ ,

$$U(x+s, y+t) - U(x,y) = [U(x+s, y+t) - U(x, y+t)] + [U(x, y+t) - U(x,y)] \quad (*)$$

By the Mean-Value Theorem for variables  $s$  and  $t$ , respectively,

$$U(x+s, y+t) - U(x, y+t) = U_x(x+s_1, y+t) \cdot s \text{ for some } |s_1| < |s|$$

$$\text{and } U(x, y+t) - U(x,y) = U_y(x, y+t_1) \cdot t \text{ for some } |t_1| < |t|$$

Let  $Q(s,t) = [U(x+s, y+t) - U(x,y)] - [U_x(x,y) \cdot s + U_y(x,y) \cdot t]$ . Now, (\*) gives

$$\frac{Q(s,t)}{s+it} = \frac{s}{s+it} \cdot [U_x(x+s_1, y+t) - U_x(x,y)] + \frac{t}{s+it} \cdot [U_y(x, y+t_1) - U_y(x,y)]$$

But,  $|s_1| < |s| \leq |s+it|$  and  $|t_1| < |t| \leq |s+it|$ , so letting  $s+it \rightarrow 0$ , then

$$\frac{Q(s,t)}{s+it} \rightarrow 0$$

Therefore  $U(x+s, y+t) - U(x,y) = U_x(x,y)s + U_y(x,y)t + Q(s,t)$

and similarly  $V(x+s, y+t) - V(x,y) = V_x(x,y)s + V_y(x,y)t + R(s,t)$  where  $\lim_{s+it \rightarrow 0} \frac{R(s,t)}{s+it} = 0$ .

$$\text{Now, } \frac{f(z+s+it) - f(z)}{s+it} = U_x(z) + iV_x(z) + \frac{Q(s,t) + iR(s,t)}{s+it} \rightarrow U_x(z) + iV_x(z).$$

as  $s+it \rightarrow 0$ .

Thm. Let  $G$  be either  $\mathbb{C}$  or some open disk

If  $u: G \rightarrow \mathbb{R}$  is a harmonic, then  $u$  has a harmonic conjugate.

There is, there is  $v: G \rightarrow \mathbb{R}$  s.t.  $u+iv$  is analytic in  $G$ .

Proof. Let  $G = B_R(0)$ ,  $0 < R \leq \infty$ . For a harmonic  $u: G \rightarrow \mathbb{R}$ , define

$$v(x,y) = \int_0^y u_x(x,t) dt + \varphi(x).$$

where  $\varphi(x)$  s.t.  $v_x = -u_y$ . Now, differentiating both sides with respect to  $x$ :

$$v_x(x,y) = \int_0^y u_{xx}(x,t) dt + \varphi'(x)$$

$$= \int_0^y -u_{xy}(x,t) dt + \varphi'(x)$$

$$= -u_y(x,y) + u_y(x,0) + \varphi'(x)$$

So,  $\varphi'(x) = -u_y(x,0)$ . Now,

$$v(x,y) = \int_0^y u_x(x,t) dt - \int_0^x u_y(s,0) ds$$

satisfies the Cauchy-Riemann equation.

Zeros of Analytic functions.

Def. If  $f: G \rightarrow \mathbb{C}$  is analytic and  $a \in G$  s.t.  $f(a) = 0$ ,  
then  $a$  is called a zero of  $f$  of multiplicity  $m \geq 1$  if:

there exists an analytic  $g: G \rightarrow \mathbb{C}$  s.t.  $f(z) = (z-a)^m g(z)$ , where  $g(a) \neq 0$ .

Index of a closed Curve.

Let  $\gamma(t) = a + e^{2\pi i n t}$  ( $0 \leq t \leq 1$ ). Then,

$$\int_{\gamma} \frac{1}{z-a} dz = \int_0^1 e^{-2\pi i n t} \cdot 2\pi i n \cdot e^{2\pi i n t} dt = 2\pi n \cdot i.$$

In general,

Lemma. Let  $\gamma: [0,1] \rightarrow \mathbb{C}$  be a closed rectifiable curve and  $a \notin \text{Im } \gamma$ , then

$$\frac{1}{2\pi i} \cdot \int_{\gamma} \frac{1}{z-a} dz$$

is an 'integer'.

Proof. Assume that  $\gamma$  is a smooth map. Define  $g: [0,1] \rightarrow \mathbb{C}$  as

$$g(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s)-a} ds$$

Then,  $g(0) = 0$  and  $g(1) = \int_0^1 \frac{\gamma'(s)}{\gamma(s)-a} ds = \int_{\gamma} \frac{1}{z-a} dz$

And  $g(t) = \frac{\gamma(t)}{\gamma(t)-a}$  for  $0 \leq t \leq 1$ .

But,  $\frac{d}{dt} e^{-g(t)} \cdot (\gamma(t)-a) = e^{-g(t)} \cdot \gamma'(t) - g'(t) \cdot e^{-g(t)} \cdot (\gamma(t)-a) = 0$

So,  $e^{-g(t)}(\gamma(t)-a)$  is the constant,

$$e^{-g(0)}(\gamma(0)-a) = \gamma(0)-a = e^{-g(1)}(\gamma(1)-a)$$

Since  $\gamma$  is closed curve, so  $\gamma(1) = \gamma(0)$ ,

$$e^{-g(1)} = 1 \quad \text{or} \quad g(1) = 2k\pi \cdot i \quad \text{for some } k \in \mathbb{Z}.$$

Def. If  $\gamma$  is a closed rectifiable curve in  $\mathbb{C}$  and  $a \notin \text{Im } \gamma$ ,

$$n(\gamma; a) \stackrel{\text{def}}{=} \frac{1}{2\pi i} \cdot \int_{\gamma} \frac{1}{z-a} dz$$

is called the index of  $\gamma$  with respect to  $a$ .

Morera Thm.

Thm. Let  $f: G \rightarrow \mathbb{C}$  be a continuous function on a region  $G$  such that

$$\int_T f = 0$$

for every triangular path  $T$  in  $G$ , then  $f$  is analytic in  $G$ .

Proof. Enough to show that  $f$  has a primitive wlog, let  $G = B_R(a)$ . Define

$$F(z) = \int_{[\alpha, z]} f(w) dw, \text{ where } [\alpha, z]: (1-t)\alpha + tz, \quad 0 \leq t \leq 1.$$

Let  $z_0 \in G$  be fixed. Then,  $F(z) = \int_{[\alpha, z_0]} f + \int_{[z_0, z]} f$ , so

$$\frac{F(z) - F(z_0)}{z - z_0} = \frac{1}{z - z_0} \cdot \int_{[z_0, z]} f$$

$$\begin{aligned} \text{Therefore, } \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) &= \frac{1}{z - z_0} \cdot \int_{[z_0, z]} f - f(z_0) \\ &= \frac{1}{z - z_0} \cdot \int_{[z_0, z]} [f(w) - f(z_0)] dw. \end{aligned}$$

$$\text{Now, } \left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| = \left| \frac{1}{z - z_0} \cdot \int_{[z_0, z]} [f(w) - f(z_0)] dw \right| \leq |f(z) - f(z_0)| \rightarrow 0.$$

